

CHAPTER 2 – SATELLITE DATA PROCESSING

2.1 Introduction

Satellite data processing is a fundamental component of remote sensing and geospatial analysis. Modern Earth observation satellites continuously acquire vast amounts of information about the Earth's surface, atmosphere, oceans, and environmental conditions. These datasets are processed and analyzed to generate meaningful information for scientific research, natural resource management, urban planning, agriculture, forestry, disaster management, and environmental monitoring.

Satellite imagery consists of digital representations of the Earth's surface captured by sensors mounted on satellites. These sensors record electromagnetic radiation reflected or emitted by various surface features. The collected information is transmitted to ground stations, processed using specialized software, and transformed into useful products such as land use maps, vegetation indices, and classified images.

With advancements in satellite technology, high-resolution multispectral and hyperspectral datasets have become widely available. Modern satellite missions provide regular and repetitive observations, enabling the monitoring of environmental changes over time. Satellite data processing involves several stages including data acquisition, preprocessing, enhancement, analysis, classification, and interpretation.

In this study, satellite data processing forms the foundation for image classification and vegetation analysis. Landsat 9 satellite imagery is utilized to perform multispectral image analysis, NDVI computation, and machine learning-based image classification. Therefore, understanding satellite missions, sensor characteristics, spectral bands, and data acquisition procedures is essential before conducting practical analysis.

2.1.1 Overview of Satellite Data

Satellite data refers to information collected by sensors mounted on Earth-observing satellites. These sensors capture electromagnetic radiation reflected or emitted from the Earth's surface in various spectral regions. The recorded data are stored in digital form and can be analyzed using Geographic Information Systems (GIS), remote sensing software, and programming tools.

Satellite data are generally available in raster format, where the image is composed of individual pixels. Each pixel represents a specific ground area and contains numerical values corresponding to the amount of reflected energy detected by the sensor. These values can be used to identify different land cover features such as vegetation, water bodies, urban areas, barren land, and mining regions.

2.1.2 Importance of Satellite Imagery in Remote Sensing

Satellite imagery plays a significant role in remote sensing because it provides continuous, large-scale, and repetitive observations of the Earth's surface. Unlike conventional field surveys, satellite observations can cover extensive geographic regions within a short period of time.

The importance of satellite imagery includes:

- Monitoring environmental changes.
- Land use and land cover mapping.
- Agricultural assessment.
- Forest resource monitoring.
- Water resource management.
- Disaster management and emergency response.
- Climate and weather studies.
- Urban planning and infrastructure development.

Satellite imagery enables decision-makers and researchers to obtain accurate spatial information that supports planning, management, and policy formulation.

2.1.3 Role of Satellite Data in Earth Observation

Earth observation refers to the systematic collection of information about the physical, biological, and human environments of the planet. Satellite data serve as one of the primary sources of Earth observation information.

Modern Earth observation systems monitor:

- Vegetation health.
- Water resources.
- Agricultural activities.
- Urban expansion.
- Forest cover changes.
- Climate variability.
- Natural disasters.

Satellite observations provide valuable information at local, regional, national, and global scales. The integration of remote sensing data with GIS and machine learning techniques has significantly enhanced Earth observation capabilities.

2.1.4 Advantages of Satellite Imagery

Satellite imagery offers several advantages compared to traditional surveying methods.

Major advantages include:

- Large area coverage.
- Repetitive observations.
- Cost-effective monitoring.

- Availability of historical datasets.
- Access to remote and inaccessible regions.
- Digital data suitable for computer processing.
- Support for automated image classification.

These advantages make satellite imagery one of the most important sources of geospatial information.

2.1.5 Difference Between Aerial Photography and Satellite Imagery

Although both aerial photography and satellite imagery are used for observing the Earth's surface, they differ in terms of platform, coverage, resolution, and applications.

Parameter	Aerial Photography	Satellite Imagery
Platform	Aircraft	Satellite
Coverage Area	Small	Large
Revisit Capability	Limited	Regular
Cost	High	Relatively Low
Data Format	Photographs	Digital Raster Data
Applications	Local Mapping	Regional and Global Monitoring

Satellite imagery is generally preferred for large-scale environmental monitoring because of its extensive coverage and continuous availability.

2.2 Indian Remote Sensing Satellite Programme

India has developed one of the largest and most successful civilian remote sensing programmes in the world. The Indian Remote Sensing (IRS) Programme was initiated by the Indian Space Research Organisation (ISRO) to provide reliable satellite-based information for natural resource management and environmental monitoring.

The IRS programme has significantly contributed to agriculture, forestry, water resources, geology, urban planning, disaster management, and environmental assessment. Data acquired from Indian satellites are widely used by government organizations, research institutions, universities, and industries.

The National Remote Sensing Centre (NRSC), operating under ISRO, is responsible for the acquisition, processing, distribution, and management of remote sensing data in India.

2.2.1 Introduction to Indian Remote Sensing (IRS)

The Indian Remote Sensing Programme was established to develop indigenous Earth observation capabilities and reduce dependence on foreign satellite systems. Over the years, India has launched several remote sensing satellites equipped with advanced sensors capable of collecting high-quality spatial data.

The IRS programme is considered one of the largest civilian remote sensing programmes globally and has gained international recognition for its operational capabilities and data quality.

2.2.1.1 History of IRS Programme

The foundation of India's remote sensing programme was laid during the 1970s. Early experimental missions demonstrated the potential of satellite technology for resource monitoring and national development.

A major milestone was achieved with the launch of IRS-1A on March 17, 1988. It became India's first operational remote sensing satellite and marked the beginning of systematic Earth observation activities.

Following the success of IRS-1A, several advanced satellite missions were launched, including:

- IRS-1B
- IRS-P Series
- Resourcesat Series
- Cartosat Series
- Oceansat Series
- RISAT Series

These missions have strengthened India's capabilities in satellite-based Earth observation and geospatial applications.

2.2.1.2 Objectives of IRS Programme

The primary objectives of the IRS programme include:

- Natural resource inventory and management.
- Agricultural monitoring.
- Water resource assessment.
- Forest monitoring.
- Urban and regional planning.
- Environmental monitoring.
- Disaster management support.
- Sustainable development planning.

The programme aims to provide reliable spatial information for informed decision-making and national development.

2.2.1.3 Applications of IRS Satellites

IRS satellites support numerous applications across different sectors.

Major applications include:

- Crop acreage estimation.
- Crop health monitoring.
- Drought assessment.
- Watershed management.
- Groundwater prospect mapping.
- Forest inventory studies.
- Coastal zone management.
- Urban growth analysis.
- Disaster management and damage assessment.

The availability of IRS data has significantly improved the efficiency and accuracy of resource management activities throughout India.

2.2.2 IRS-1A and IRS Series

The IRS (Indian Remote Sensing) satellite series represents one of the most important achievements of the Indian Space Research Organisation (ISRO). The series was developed to provide reliable Earth observation data for natural resource management, environmental monitoring, and national development. IRS satellites have played a significant role in establishing India's self-reliance in remote sensing technology.

2.2.2.1 Launch and Development

IRS-1A was launched on 17 March 1988 from the Baikonur Cosmodrome, Kazakhstan. It was India's first operational remote sensing satellite and marked the beginning of a systematic Earth observation programme in the country.

The success of IRS-1A was followed by several subsequent missions including IRS-1B, IRS-1C, IRS-1D, and the IRS-P series. These satellites were equipped with increasingly advanced sensors capable of acquiring high-resolution imagery for various applications.

The IRS programme significantly enhanced India's ability to monitor natural resources, support agricultural planning, and conduct environmental assessments.

2.2.2.2 Sensor Characteristics

IRS satellites carried Linear Imaging Self-Scanning Sensors (LISS) designed to capture multispectral imagery. These sensors recorded reflected solar radiation in multiple spectral bands, enabling the identification and classification of various land cover features.

Major sensors used in the IRS programme include:

- LISS-I
- LISS-II
- LISS-III

- LISS-IV
- PAN Camera
- WiFS Sensor
- AWiFS Sensor

These sensors provide varying levels of spatial and spectral resolution suitable for different applications.

2.2.2.3 Applications

IRS satellites have been extensively used in:

- Land use and land cover mapping.
- Agricultural monitoring.
- Watershed management.
- Forest resource assessment.
- Groundwater exploration.
- Urban planning.
- Disaster management.
- Environmental impact studies.

The long-term availability of IRS data has made it one of the most valuable sources of geospatial information in India.

2.2.3 Resourcesat Series

The Resourcesat series was developed by ISRO to support agricultural monitoring, natural resource management, and environmental studies. These satellites provide high-quality multispectral imagery and have become an important source of data for remote sensing applications in India.

Resourcesat satellites are considered successors to the IRS programme and offer improved sensor capabilities and data quality.

2.2.3.1 Introduction

Resourcesat-1 was launched in 2003 with the objective of enhancing resource monitoring capabilities. It carried advanced imaging sensors capable of capturing detailed information about vegetation, land use, water resources, and environmental conditions.

The mission was followed by Resourcesat-2 and Resourcesat-2A, which provided improved continuity of Earth observation services.

2.2.3.2 LISS Sensors

One of the most important features of the Resourcesat series is the use of LISS (Linear Imaging Self Scanning Sensor) instruments.

Major sensors include:

LISS-III

Characteristics:

- Spatial Resolution: 23.5 m
- Multispectral imaging
- Suitable for vegetation and land use studies

LISS-IV

Characteristics:

- Spatial Resolution: 5.8 m
- High-resolution imagery
- Urban and infrastructure mapping

AWiFS

Characteristics:

- Spatial Resolution: 56 m
- Wide area coverage
- Agricultural monitoring

The LISS-III sensor is particularly important because it is frequently used for vegetation assessment and land cover classification studies.

2.2.3.3 Applications in Agriculture

Resourcesat data are widely used in agricultural applications including:

- Crop acreage estimation.
- Crop health monitoring.
- Yield forecasting.
- Drought assessment.
- Irrigation management.
- Precision agriculture.

The multispectral capabilities of Resourcesat satellites make them highly effective for vegetation analysis and resource management.

2.2.4 Cartosat Series

The Cartosat series was developed to provide high-resolution imagery for cartographic and mapping applications. These satellites are designed to support urban planning, infrastructure development, land administration, and geographic information systems.

2.2.4.1 Introduction

Cartosat satellites represent India's advanced mapping missions. They provide high spatial resolution imagery suitable for detailed topographic mapping and urban analysis.

Several Cartosat missions have been launched, including:

- Cartosat-1
- Cartosat-2
- Cartosat-2A
- Cartosat-2B
- Cartosat-2C
- Cartosat-3

These missions have significantly enhanced India's geospatial capabilities.

2.2.4.2 High Resolution Mapping

The primary objective of the Cartosat series is the generation of high-resolution spatial data.

Applications include:

- Topographic mapping.
- Digital elevation model generation.
- Urban infrastructure planning.
- Transportation network analysis.
- Smart city development.

Cartosat imagery provides detailed spatial information that supports accurate mapping and planning activities.

2.2.4.3 Applications

Major applications of Cartosat satellites include:

- Urban growth monitoring.
- Infrastructure development.
- Land administration.
- Transportation planning.
- Defense and security applications.

- Disaster management.

The high spatial resolution of Cartosat imagery makes it particularly valuable for detailed mapping projects.

2.2.5 Oceansat Series

The Oceansat series was developed by ISRO for oceanographic and atmospheric studies. These satellites collect information about sea surface conditions, ocean productivity, weather systems, and climate processes.

2.2.5.1 Introduction

Oceansat satellites are dedicated Earth observation missions designed to monitor marine and atmospheric environments. The data obtained from these satellites support fisheries management, weather forecasting, and climate research.

Major missions include:

- Oceansat-1
- Oceansat-2
- Oceansat-3

2.2.5.2 Ocean Monitoring Applications

Oceansat data are used for:

- Sea surface temperature monitoring.
- Ocean colour analysis.
- Chlorophyll concentration estimation.
- Marine ecosystem assessment.
- Coastal zone management.

These applications contribute to sustainable utilization of marine resources.

2.2.5.3 Climate Studies

Satellite observations from Oceansat missions provide valuable information for understanding climate variability and atmospheric processes.

Applications include:

- Monsoon monitoring.
- Cyclone tracking.
- Climate change studies.
- Weather forecasting.
- Ocean-atmosphere interaction analysis.

The continuous observation capability of Oceansat satellites supports scientific research and operational forecasting activities.

Table 2.1 Comparison of Major Indian Remote Sensing Satellites

Satellite	Launch Purpose	Main Application
IRS Series	Earth Observation	Resource Monitoring
Resourcesat	Agriculture & Resources	Crop Monitoring
Cartosat	High Resolution Mapping	Urban Planning
Oceansat	Ocean Observation	Marine Studies
RISAT	Radar Imaging	Disaster Monitoring

The Indian Remote Sensing programme has successfully established India as a global leader in satellite-based Earth observation. The various satellite missions provide complementary datasets that support resource management, environmental monitoring, scientific research, and national development.

2.3 International Earth Observation Satellites

Remote sensing has become a global technology supported by numerous Earth observation missions operated by different countries and international space agencies. These satellites provide valuable information for environmental monitoring, agriculture, forestry, water resources, climate studies, disaster management, and urban planning.

Among the most important international Earth observation missions are the Landsat Programme, Sentinel Programme, and MODIS missions. These satellite systems provide freely accessible data that are widely used by researchers, government organizations, and industries around the world.

The present study primarily utilizes Landsat satellite imagery; therefore, understanding the characteristics and applications of international satellite missions is essential.

2.3.1 Landsat Programme

The Landsat Programme is the world's longest-running Earth observation satellite mission. It is jointly managed by the National Aeronautics and Space Administration (NASA) and the United States Geological Survey (USGS).

Since its inception, Landsat has continuously provided valuable information about land cover, vegetation, water resources, urban development, and environmental change.

2.3.1.1 History of Landsat

The Landsat programme began with the launch of Landsat-1 on July 23, 1972. It was the first satellite specifically designed for monitoring the Earth's land resources.

The success of Landsat-1 led to a series of subsequent missions, resulting in one of the most valuable archives of Earth observation data available today.

Major Landsat missions include:

- Landsat-1 (1972)
- Landsat-2 (1975)
- Landsat-3 (1978)
- Landsat-4 (1982)
- Landsat-5 (1984)
- Landsat-7 (1999)
- Landsat-8 (2013)
- Landsat-9 (2021)

These missions have provided continuous observations of the Earth's surface for more than five decades.

2.3.1.2 Evolution from Landsat 1 to Landsat 9

The Landsat programme has undergone significant technological improvements over time.

Early Landsat missions utilized Multispectral Scanner (MSS) sensors with limited spatial and spectral resolution. Later missions introduced advanced sensors such as:

- Thematic Mapper (TM)
- Enhanced Thematic Mapper Plus (ETM+)
- Operational Land Imager (OLI)
- Thermal Infrared Sensor (TIRS)

These advancements improved image quality, spectral coverage, radiometric resolution, and data accuracy.

Landsat-9 currently represents the latest generation of the programme and continues to provide high-quality multispectral imagery for scientific and operational applications.

2.3.1.3 Applications

Landsat imagery is extensively used in:

- Land use and land cover mapping.
- Vegetation monitoring.
- Agricultural assessment.
- Water resource management.
- Urban growth analysis.

- Forest monitoring.
- Environmental assessment.
- Climate studies.
- Disaster management.

The free availability of Landsat data has made it one of the most widely used satellite datasets in remote sensing research.

2.3.2 Sentinel Programme

The Sentinel Programme is a family of Earth observation satellites developed by the European Space Agency (ESA) under the Copernicus Programme. The mission was established to provide continuous environmental monitoring and support sustainable development initiatives.

Sentinel satellites offer high-quality optical, radar, atmospheric, and oceanographic observations that complement data acquired from other satellite missions.

2.3.2.1 Sentinel-1

Sentinel-1 is a radar-based Earth observation mission that utilizes Synthetic Aperture Radar (SAR) technology.

Key characteristics include:

- Active remote sensing system.
- Day and night operation.
- All-weather imaging capability.
- Cloud penetration capability.

Applications include:

- Flood monitoring.
- Disaster management.
- Land deformation analysis.
- Agricultural monitoring.
- Forest assessment.

Because Sentinel-1 uses radar technology, it can acquire data even under cloudy conditions where optical satellites may fail.

2.3.2.2 Sentinel-2

Sentinel-2 is an optical Earth observation mission designed for land monitoring applications.

The mission consists of two satellites:

- Sentinel-2A

- Sentinel-2B

Key features include:

- High spatial resolution imagery.
- Thirteen spectral bands.
- Frequent revisit capability.
- Wide area coverage.

Sentinel-2 imagery is extensively used for vegetation analysis and land cover mapping.

2.3.2.3 Applications

Sentinel missions support a wide range of applications including:

- Agricultural monitoring.
- Forest resource assessment.
- Land cover classification.
- Water quality analysis.
- Environmental monitoring.
- Disaster response.
- Urban planning.

The high temporal resolution of Sentinel data makes it suitable for monitoring rapidly changing environmental conditions.

2.3.3 MODIS

MODIS (Moderate Resolution Imaging Spectroradiometer) is an important Earth observation instrument carried onboard NASA's Terra and Aqua satellites.

MODIS was designed to monitor global environmental processes and provide continuous observations of the Earth's atmosphere, land, and oceans.

2.3.3.1 Terra Mission

The Terra satellite was launched by NASA in December 1999 as part of the Earth Observing System (EOS).

The MODIS instrument onboard Terra collects data related to:

- Vegetation dynamics.
- Land surface temperature.
- Snow cover.
- Forest fires.
- Atmospheric conditions.

Terra provides valuable information for environmental and climate research.

2.3.3.2 Aqua Mission

The Aqua satellite was launched in May 2002 and focuses primarily on monitoring the Earth's water cycle.

The mission observes:

- Ocean processes.
- Water vapour distribution.
- Cloud formation.
- Precipitation patterns.
- Surface water resources.

Together, Terra and Aqua provide comprehensive observations of global environmental systems.

2.3.3.3 Applications

MODIS data are widely used in:

- Climate change studies.
- Drought monitoring.
- Vegetation analysis.
- Forest fire detection.
- Water resource assessment.
- Global environmental monitoring.

Although MODIS has lower spatial resolution compared to Landsat and Sentinel, its high temporal frequency makes it extremely valuable for large-scale monitoring applications.

2.3.4 Comparison of Major Satellite Missions

Different Earth observation missions are designed for different objectives. Therefore, selecting an appropriate satellite depends on the requirements of a specific application.

Table 2.2 Comparison of Landsat, Sentinel and MODIS

Parameter	Landsat 9	Sentinel-2	MODIS
Agency	NASA/USGS	ESA	NASA
Launch Year	2021	2015	1999/2002
Spatial Resolution	30 m	10–20 m	250–1000 m
Spectral Bands	11	13	36
Revisit Period	16 Days	5 Days	1–2 Days
Major Application	Land Monitoring	Vegetation Analysis	Climate Monitoring

Among these missions, Landsat and Sentinel are the most widely used datasets for land cover classification and vegetation analysis. In the present study, Landsat 9 imagery is selected due to its suitable spatial resolution, multispectral capabilities, and free availability.

The understanding of international satellite missions provides the necessary background for studying Landsat 9 in greater detail, which forms the primary data source for the subsequent image processing and classification tasks.

2.4 Landsat 9 Satellite

Landsat 9 is the latest satellite in the Landsat Earth observation programme jointly operated by the National Aeronautics and Space Administration (NASA) and the United States Geological Survey (USGS). It continues the long-standing mission of monitoring the Earth's land surface and provides high-quality multispectral imagery for scientific research, environmental monitoring, agriculture, forestry, water resource management, and urban planning.

Landsat 9 ensures the continuity of the global Earth observation record established by previous Landsat missions. The satellite acquires imagery with improved radiometric performance and provides consistent observations that support long-term environmental studies.

In the present study, Landsat 9 imagery serves as the primary data source for RGB composite generation, vegetation analysis using NDVI, and machine learning-based image classification.

2.4.1 Introduction to Landsat 9

Landsat 9 was launched on 27 September 2021 to continue the legacy of the Landsat programme. The satellite operates in a near-polar, sun-synchronous orbit, allowing it to capture images of the Earth's surface under consistent illumination conditions.

The mission was developed to maintain uninterrupted Earth observation data collection and to support various scientific and operational applications worldwide.

2.4.1.1 Launch Details

The satellite was launched aboard an Atlas V rocket from Vandenberg Space Force Base in California, United States.

Key specifications of Landsat 9 include:

Parameter	Value
Launch Date	27 September 2021
Orbit Type	Sun-Synchronous Orbit
Altitude	Approximately 705 km
Revisit Time	16 Days
Mission Type	Earth Observation

Landsat 9 follows the same orbital path as Landsat 8, ensuring consistent global coverage and long-term data continuity.

2.4.1.2 Mission Objectives

The major objectives of the Landsat 9 mission include:

- Monitoring global land cover changes.
- Supporting agricultural management.
- Assessing vegetation conditions.
- Monitoring water resources.
- Tracking environmental changes.
- Supporting climate studies.
- Providing reliable Earth observation data for scientific research.

The mission contributes significantly to sustainable resource management and environmental monitoring at regional and global scales.

2.4.2 Sensors on Landsat 9

Landsat 9 carries two advanced imaging instruments designed to acquire high-quality multispectral and thermal imagery.

These sensors are:

1. Operational Land Imager-2 (OLI-2)
2. Thermal Infrared Sensor-2 (TIRS-2)

Together, these instruments provide comprehensive spectral coverage for Earth observation applications.

2.4.2.1 Operational Land Imager-2 (OLI-2)

The Operational Land Imager-2 (OLI-2) is the primary multispectral imaging instrument onboard Landsat 9.

It captures reflected solar radiation in visible, near-infrared, and shortwave infrared portions of the electromagnetic spectrum.

Major features include:

- High radiometric sensitivity.
- Multispectral imaging capability.
- Improved signal-to-noise ratio.
- Enhanced image quality.

OLI-2 is responsible for collecting most of the data used in vegetation monitoring, land cover classification, and environmental assessment studies.

2.4.2.2 Thermal Infrared Sensor-2 (TIRS-2)

The Thermal Infrared Sensor-2 (TIRS-2) acquires thermal information emitted by the Earth's surface.

Applications include:

- Surface temperature estimation.
- Drought assessment.
- Water resource monitoring.
- Urban heat island studies.
- Climate research.

The combination of OLI-2 and TIRS-2 allows Landsat 9 to provide both spectral and thermal information about the Earth's surface.

2.4.3 Spectral Bands of Landsat 9

One of the most important characteristics of Landsat 9 is its multispectral imaging capability. The satellite records information in multiple spectral bands, each designed for specific observation purposes.

These bands allow the identification and classification of different land cover features based on their spectral responses.

Table 2.3 Spectral Bands of Landsat 9

Band	Name	Wavelength (µm)	Resolution
Band 1	Coastal Aerosol	0.43–0.45	30 m
Band 2	Blue	0.45–0.51	30 m
Band 3	Green	0.53–0.59	30 m
Band 4	Red	0.64–0.67	30 m
Band 5	Near Infrared (NIR)	0.85–0.88	30 m
Band 6	SWIR-1	1.57–1.65	30 m
Band 7	SWIR-2	2.11–2.29	30 m
Band 8	Panchromatic	0.50–0.68	15 m
Band 9	Cirrus	1.36–1.38	30 m
Band 10	Thermal Infrared-1	10.6–11.2	100 m
Band 11	Thermal Infrared-2	11.5–12.5	100 m

2.4.3.1 Coastal Aerosol Band

Band 1 is primarily used for coastal and atmospheric studies. It assists in detecting fine particles and atmospheric aerosols.

2.4.3.2 Blue Band

Band 2 is useful for:

- Water body mapping.
- Soil discrimination.
- Coastal studies.
- Bathymetric analysis.

2.4.3.3 Green Band

Band 3 represents green vegetation reflectance and is useful for vegetation studies and land cover mapping.

2.4.3.4 Red Band

Band 4 is strongly absorbed by healthy vegetation and is important for vegetation analysis.

Applications include:

- Land cover classification.
- Vegetation assessment.
- Agricultural studies.

2.4.3.5 Near Infrared (NIR) Band

Band 5 is one of the most important bands for vegetation analysis.

Healthy vegetation reflects a large amount of near-infrared radiation, making this band highly useful for:

- Vegetation monitoring.
- Crop health assessment.
- Forest studies.
- NDVI calculation.

The present study extensively uses Band 5 for vegetation analysis.

2.4.3.6 SWIR-1 Band

Band 6 is sensitive to vegetation moisture and soil moisture conditions.

Applications include:

- Drought assessment.
- Vegetation moisture analysis.

- Geological studies.

2.4.3.7 SWIR-2 Band

Band 7 is useful for:

- Mineral mapping.
- Burn severity assessment.
- Geological investigations.

2.4.3.8 Panchromatic Band

Band 8 provides higher spatial resolution imagery (15 m) and is commonly used for image sharpening.

2.4.3.9 Cirrus Band

Band 9 is specifically designed for cloud detection and atmospheric correction.

2.4.4 Applications of Landsat 9

Landsat 9 data are used in a wide range of remote sensing applications.

Major applications include:

- Land use and land cover mapping.
- Agricultural monitoring.
- Forest resource assessment.
- Water resource management.
- Urban growth monitoring.
- Environmental impact assessment.
- Disaster management.
- Climate studies.
- Change detection analysis.

In this project, Landsat 9 imagery is utilized to generate RGB composite images, compute the Normalized Difference Vegetation Index (NDVI), and perform machine learning-based image classification using K-Means clustering.

The multispectral capabilities of Landsat 9 make it highly suitable for identifying vegetation, water bodies, built-up areas, mining regions, and other land cover classes within the study area.

2.5 Satellite Resolution

Satellite imagery is composed of digital information acquired by sensors operating in different regions of the electromagnetic spectrum. The quality and usefulness of satellite data depend largely on the resolution of the sensor. Resolution determines the level of detail that can be observed and analyzed from satellite imagery.

In remote sensing, resolution refers to the ability of a sensor to distinguish between objects or features on the Earth's surface. Different types of resolutions provide different kinds of information, and understanding these resolutions is essential for selecting suitable satellite data for specific applications.

The four major types of resolution used in remote sensing are:

1. Spatial Resolution
2. Spectral Resolution
3. Temporal Resolution
4. Radiometric Resolution

2.5.1 Introduction to Resolution

The term "resolution" describes the capability of a remote sensing system to detect, record, and distinguish information about the Earth's surface. Higher resolution generally results in more detailed information, although it may require greater storage capacity and processing resources.

Different applications require different types of resolution. For example, urban mapping requires high spatial resolution, whereas climate studies may require high temporal resolution.

2.5.2 Spatial Resolution

Spatial resolution refers to the size of the smallest ground feature that can be represented by a single pixel in a satellite image.

In simpler terms, spatial resolution indicates how much area on the ground is represented by one image pixel.

2.5.2.1 Definition

Spatial resolution is the ground area covered by an individual pixel of a satellite image.

For example:

- A 30-meter resolution image means one pixel represents an area of $30\text{ m} \times 30\text{ m}$ on the ground.
- A 10-meter resolution image means one pixel represents an area of $10\text{ m} \times 10\text{ m}$.

Higher spatial resolution results in more detailed imagery.

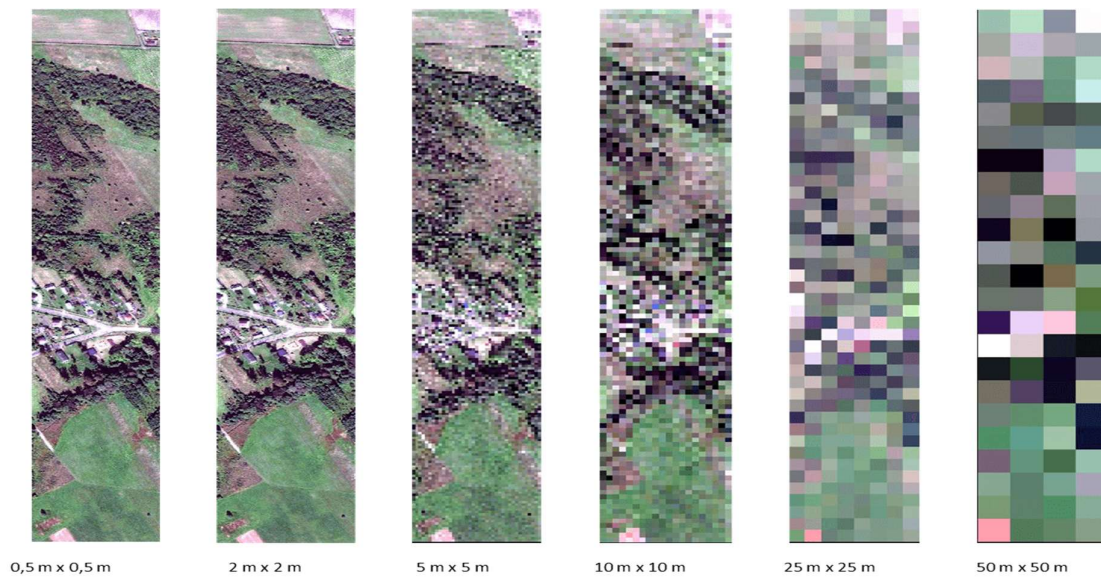


Figure 2-1: Spatial Resolution Comparison

2.5.2.2 Examples

Satellite	Spatial Resolution
Landsat 9	30 m
Sentinel-2	10 m
Cartosat-3	<1 m
MODIS	250–1000 m

Applications of high spatial resolution include:

- Urban mapping.
- Infrastructure monitoring.
- Road extraction.
- Detailed land use mapping.

Applications of moderate spatial resolution include:

- Vegetation monitoring.
- Agricultural studies.
- Environmental assessment.

In this project, Landsat 9 imagery with 30-meter spatial resolution is used for land cover classification and vegetation analysis.

2.5.3 Spectral Resolution

Spectral resolution refers to the ability of a sensor to distinguish different wavelengths of electromagnetic radiation.

Different materials reflect energy differently at various wavelengths. Therefore, multiple spectral bands enable the identification and classification of different land cover types.

2.5.3.1 Definition

Spectral resolution is defined as the number and width of spectral bands recorded by a sensor.

A sensor with more spectral bands generally provides better capability for distinguishing between different surface materials.

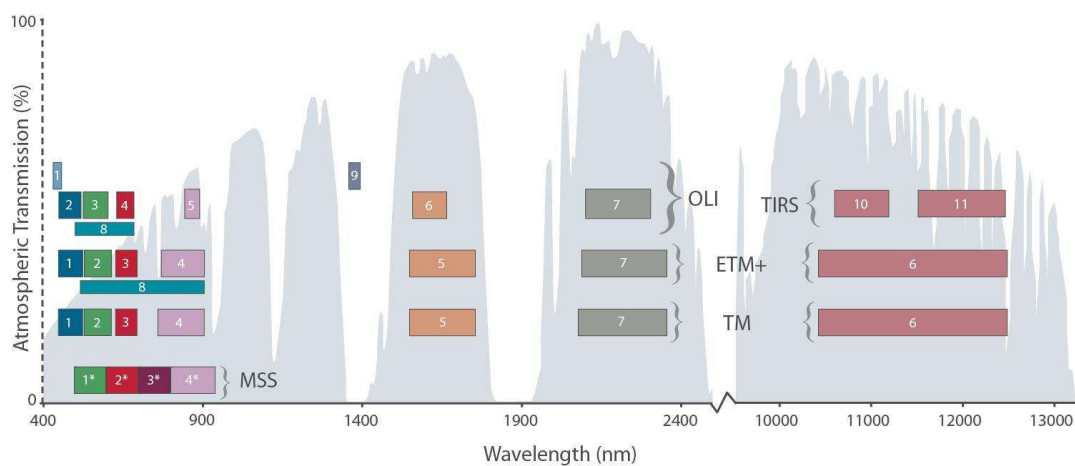


Figure 2-2: Spectral bands of Landsat 9 and their corresponding wavelength regions in the electromagnetic spectrum

2.5.3.2 Examples

Examples of spectral bands include:

- Blue Band
- Green Band
- Red Band
- Near Infrared (NIR)
- Short Wave Infrared (SWIR)

Landsat 9 records information in eleven spectral bands, providing valuable data for vegetation analysis, water resource assessment, and land cover mapping.

Applications of spectral resolution include:

- Vegetation identification.
- Water body mapping.

- Soil analysis.
- Mineral exploration.
- Environmental monitoring.

Higher spectral resolution improves the accuracy of image classification and feature extraction.

2.5.4 Temporal Resolution

Temporal resolution refers to the frequency with which a satellite revisits and captures imagery of the same location on the Earth's surface.

It is particularly important for monitoring changes over time.

2.5.4.1 Definition

Temporal resolution is the time interval between successive observations of the same area by a satellite sensor.

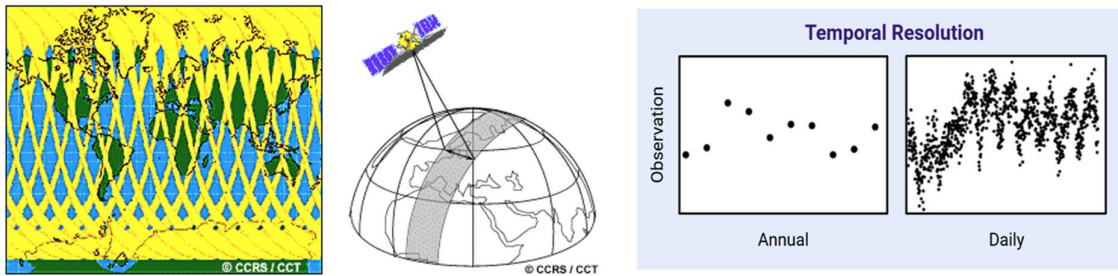


Figure 2-3: Temporal Resolution Illustration

2.5.4.2 Examples

Satellite	Revisit Time
Landsat 9	16 Days
Sentinel-2	5 Days
MODIS	1–2 Days

Applications include:

- Crop monitoring.
- Drought assessment.
- Flood monitoring.
- Forest fire detection.
- Change detection studies.

Satellites with shorter revisit periods are better suited for monitoring rapidly changing environmental conditions.

2.5.5 Radiometric Resolution

Radiometric resolution refers to the ability of a sensor to distinguish slight differences in energy intensity.

It determines the sensitivity of the sensor to variations in reflected or emitted radiation.

2.5.5.1 Definition

Radiometric resolution is the number of brightness levels that a sensor can record.

The greater the radiometric resolution, the better the sensor can detect subtle differences in surface characteristics.



Figure 2-4: Radiometric Resolution Illustration (Brightness Levels)

2.5.5.2 Examples

Common radiometric resolutions include:

Bit Depth	Number of Levels
8-bit	256
10-bit	1024
12-bit	4096
16-bit	65536

Modern satellite sensors such as Landsat 9 provide high radiometric resolution, enabling improved discrimination of land cover features.

Applications include:

- Vegetation health analysis.
- Soil moisture studies.
- Water quality assessment.
- Environmental monitoring.

2.5.6 Comparison of Different Resolutions

Each type of resolution provides unique information about the Earth's surface and contributes to the effectiveness of remote sensing applications.

Table 2.4 Comparison of Different Types of Resolution

Resolution Type	Description	Example
Spatial Resolution	Ground area represented by a pixel	30 m Landsat Pixel
Spectral Resolution	Number and width of spectral bands	11 Bands in Landsat 9
Temporal Resolution	Frequency of image acquisition	16 Days
Radiometric Resolution	Sensor sensitivity to brightness levels	16-bit Data

The selection of suitable satellite data depends on the specific objectives of a study. For land cover classification and vegetation analysis, a balance between spatial, spectral, temporal, and radiometric resolution is required.

Landsat 9 provides an effective combination of these resolutions, making it suitable for the image classification tasks performed in the present study.

2.6 Satellite Data Sources

The availability of satellite data is essential for remote sensing applications. Various national and international organizations provide access to Earth observation datasets through online geospatial portals. These portals allow researchers, government agencies, students, and industries to access satellite imagery for scientific analysis and practical applications.

For the present study, satellite data are obtained from publicly accessible platforms that provide multispectral imagery suitable for land cover classification and vegetation analysis.

The two major data sources considered in this study are:

1. USGS Earth Explorer
2. ISRO Bhuvan Geoportal

These platforms provide reliable and freely accessible satellite datasets for remote sensing research.

2.6.1 USGS Earth Explorer

USGS Earth Explorer is one of the most widely used online platforms for downloading Earth observation data. It is maintained by the United States Geological Survey (USGS) and provides access to a large archive of satellite imagery and geospatial datasets.

The platform serves as the primary source for Landsat imagery and supports users worldwide.

2.6.1.1 Introduction

Earth Explorer is a web-based application that enables users to search, visualize, and download satellite imagery based on geographic location, time period, and dataset type.

The platform provides access to historical as well as recent Earth observation data and supports a wide range of remote sensing applications.

2.6.1.2 Features

Major features of Earth Explorer include:

- Interactive map interface.
- Global data coverage.
- Multiple dataset options.
- Temporal filtering.
- Cloud cover filtering.
- Free data download.
- Metadata access.

These features make Earth Explorer a valuable tool for remote sensing studies.

2.6.1.3 Available Datasets

Earth Explorer provides access to various satellite missions including:

- Landsat Series
- Sentinel Data
- MODIS Products
- Digital Elevation Models (DEM)
- Aerial Imagery
- Climate Data

Among these datasets, Landsat imagery is the most relevant for the present study.

2.6.1.4 Importance for Landsat Data

Earth Explorer is the official platform for accessing Landsat imagery. It provides complete Landsat archives dating back to 1972 and allows users to download multispectral datasets for research and operational applications.

The Landsat 9 data used in this project were obtained through Earth Explorer.

2.6.2 ISRO Bhuvan Geoportal

Bhuvan is India's national geoportal developed and maintained by the National Remote Sensing Centre (NRSC) under the Indian Space Research Organisation (ISRO).

The platform provides geospatial data, satellite imagery, thematic maps, and various Earth observation services relevant to India.

2.6.2.1 Introduction

Bhuvan was launched to provide indigenous geospatial services and promote the use of remote sensing data in governance, research, and education.

It serves as an Indian alternative to international mapping platforms and provides access to a wide range of satellite datasets.

2.6.2.2 Features

Major features include:

- High-resolution satellite imagery.
- Thematic mapping services.
- Geospatial data visualization.
- Disaster management support.
- Resource monitoring tools.
- Web-based GIS services.

2.6.2.3 Available Datasets

Bhuvan provides access to:

- IRS Data
- Resourcesat Data
- Cartosat Data
- Oceansat Data
- Land Use Maps
- Disaster Monitoring Products

These datasets support a variety of environmental and resource management applications.

2.6.2.4 Importance for Indian Remote Sensing Data

Bhuvan serves as the primary portal for accessing Indian satellite datasets. It promotes the utilization of indigenous Earth observation resources and supports research activities related to agriculture, water resources, forestry, urban planning, and disaster management.

Both platforms are valuable sources of satellite imagery and geospatial information. In the present study, Earth Explorer is used for obtaining Landsat 9 imagery, while Bhuvan serves as a reference source for Indian remote sensing data.

2.6.3 Comparison of Data Sources

Table 2.5 Comparison of Earth Explorer and Bhuvan

Parameter	Earth Explorer	Bhuvan
Organization	USGS	ISRO
Country	USA	India
Major Dataset	Landsat	IRS Series
Coverage	Global	Primarily India
Access	Free	Free
Applications	Global Monitoring	National Applications

2.7 Data Acquisition Methodology

Data acquisition is an important stage in satellite image processing. It involves identifying a suitable study area, selecting an appropriate dataset, and downloading imagery for further analysis.

For this study, Landsat 9 imagery covering Jodhpur district was downloaded using the Earth Explorer platform.

2.7.1 Selection of Study Area

The study area selected for this project is Jodhpur district, located in the western part of Rajasthan, India.

Jodhpur is characterized by:

- Arid and semi-arid climate.
- Sparse vegetation cover.
- Urban settlements.
- Mining activities.
- Agricultural land.

These diverse land cover features make the area suitable for image classification and vegetation analysis studies.

2.7.1.1 Jodhpur District

Jodhpur is one of the largest districts in Rajasthan and plays an important role in agriculture, mining, tourism, and urban development.

2.7.1.2 Area of Interest (AOI)

The Area of Interest (AOI) was defined within Earth Explorer by locating Jodhpur on the interactive map and selecting the desired study region.

2.7.2 Data Download from USGS Earth Explorer

The Landsat 9 satellite imagery used in this study was downloaded from the USGS Earth Explorer portal. The following procedure was adopted to obtain suitable imagery for the study area.

2.7.2.1 Accessing Earth Explorer

The USGS Earth Explorer website was accessed through a web browser. A registered user account was used to log into the platform and gain access to satellite datasets available for download.

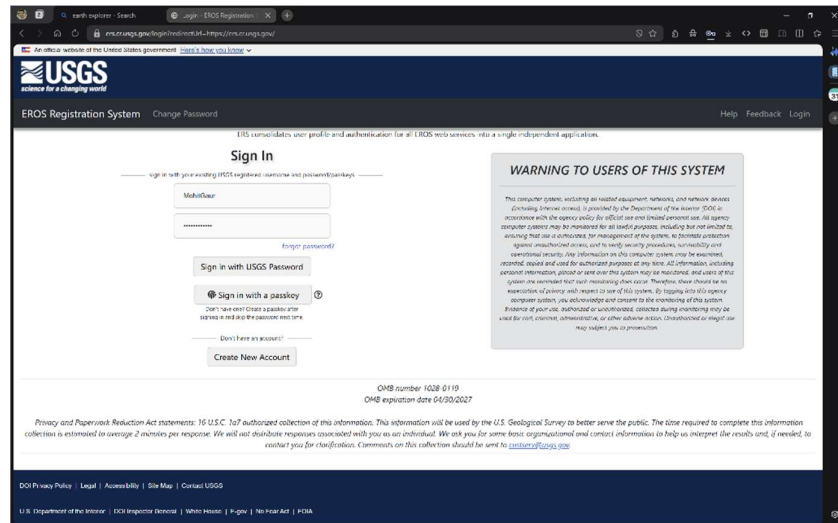


Figure 2-5: Login page of the USGS Earth Explorer platform used for accessing satellite imagery datasets

2.7.2.2 Search for Study Area

The study area, Jodhpur district of Rajasthan, India, was searched using the location search tool available on the Earth Explorer interface. The map automatically navigated to the selected location.

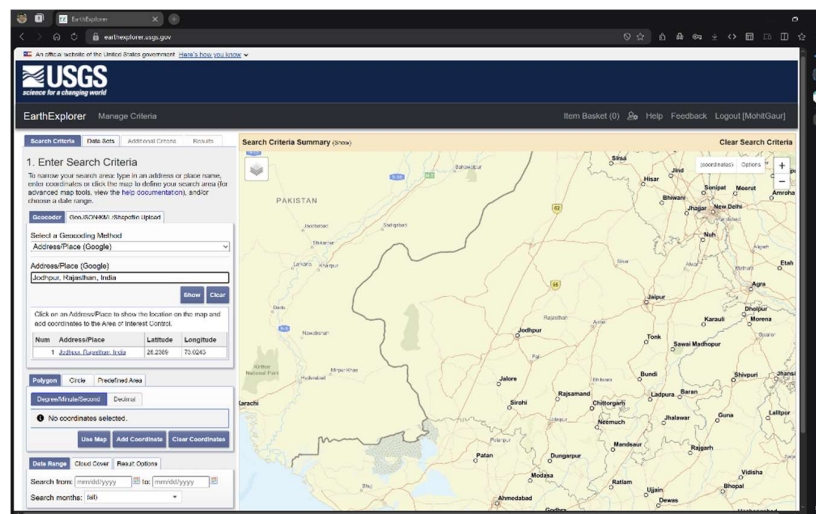


Figure 2-6: Searching and locating Jodhpur district, Rajasthan, India, using the Earth Explorer search interface

2.7.2.3 Area of Interest (AOI) Selection

An Area of Interest (AOI) was defined using the map interface. A rectangular boundary was drawn around Jodhpur and the surrounding region to ensure inclusion of major land cover features such as built-up areas, vegetation, water bodies, and mining regions.

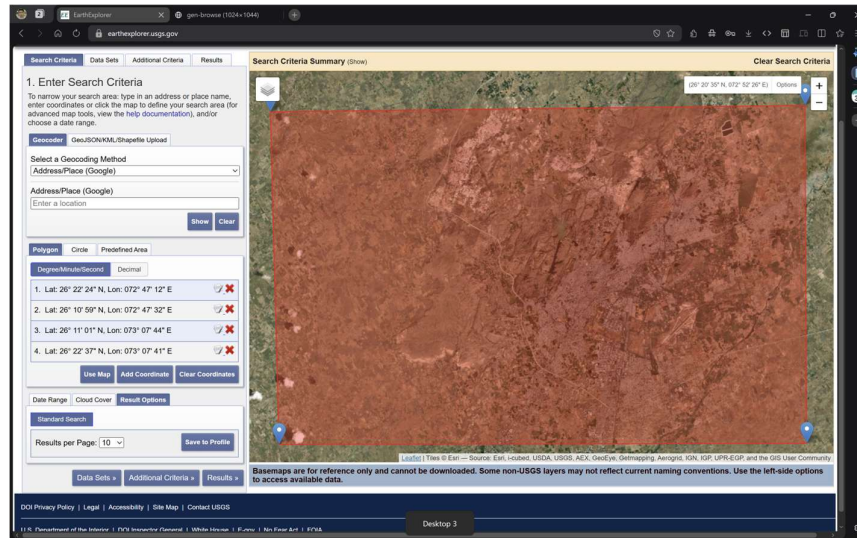


Figure 2-7: Area of Interest (AOI) covering Jodhpur and surrounding regions for satellite image analysis

2.7.2.4 Selection of Date Range

A suitable acquisition period was specified under the Date Range option. Recent imagery was preferred to ensure that the data accurately represented the current land cover conditions within the study area.

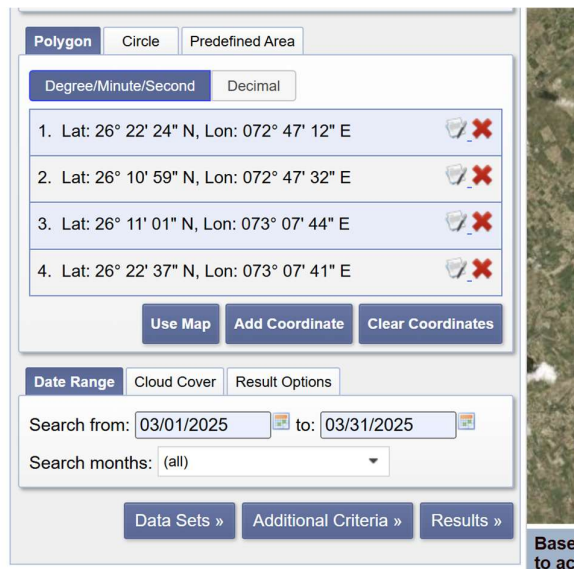


Figure 2-8: Specification of the image acquisition period for retrieving recent satellite imagery of the study area

2.7.2.5 Cloud Cover Filtering

To minimize atmospheric interference and improve image quality, a cloud cover filter of less than 10% was applied. This ensured that the selected imagery contained minimal cloud contamination.

Polygon Circle Predefined Area

Degree/Minute/Second Decimal

1. Lat: 26° 22' 24" N, Lon: 072° 47' 12" E	☁️ ❌
2. Lat: 26° 10' 59" N, Lon: 072° 47' 32" E	☁️ ❌
3. Lat: 26° 11' 01" N, Lon: 073° 07' 44" E	☁️ ❌
4. Lat: 26° 22' 37" N, Lon: 073° 07' 41" E	☁️ ❌

Use Map Add Coordinate Clear Coordinates

Date Range Cloud Cover Result Options

Cloud Cover Range: 0% - 8%

Unknown Cloud Cover Values Included ▾

This filter will only be applied to data sets that support cloud cover filtering (in the data set list denotes cloud cover support).

Data Sets > Additional Criteria > Results >

Figure 2-9: Application of cloud cover constraints to obtain high-quality cloud-free satellite imagery

2.7.2.6 Dataset Selection

The required satellite dataset was selected from the Data Sets tab. The following dataset was chosen for the study:

- Landsat Collection 2
- Landsat 9 OLI/TIRS

This dataset was selected because it provides multispectral imagery suitable for RGB composite generation, NDVI analysis, and land cover classification.

2. Select Your Data Set(s)

Check the boxes for the data set(s) you want to search. When done selecting data set(s), click the Additional Criteria or Results buttons below. Click the plus sign next to the category name to show a list of data sets.

☐ Use Data Set Prefilter (What's This?)

Data Set Search:

Basemaps are for reference only and to access available data.

Figure 2-10: Selection of Landsat Collection 2 Level-2 (Landsat 8–9 OLI/TIRS) dataset in USGS Earth Explorer

2.7.2.7 Scene Selection

After applying the search criteria, the available Landsat scenes were reviewed. The most suitable scene was selected based on:

- Cloud cover percentage
- Image quality
- Acquisition date
- Complete coverage of the study area

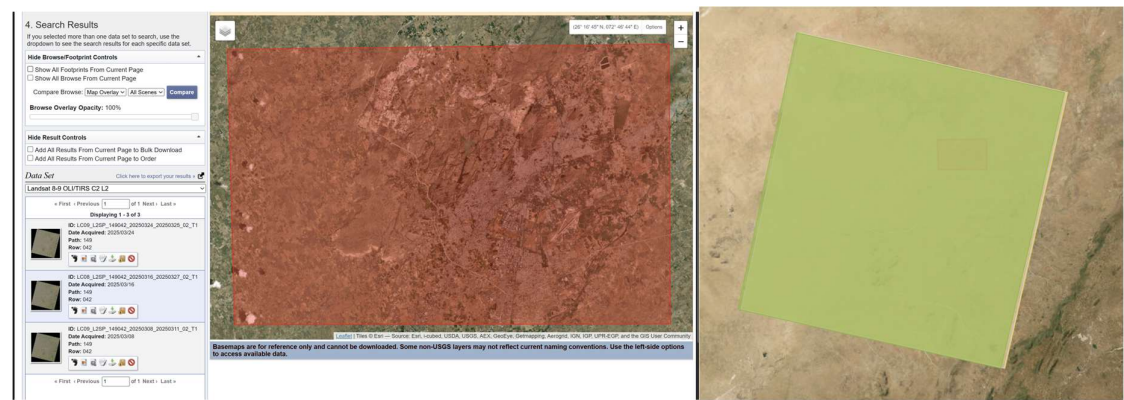


Figure 2-11: (i) Available Landsat scenes covering the selected Area of Interest
(ii) Spatial coverage of the selected Landsat scene relative to the study area



Figure 2-12: Detailed preview and metadata of the selected Landsat scene used for analysis

2.7.2.8 Downloading the Dataset

The selected Landsat 9 scene was downloaded from the download options provided by Earth Explorer. The dataset was obtained in compressed format containing multispectral band files and associated metadata.

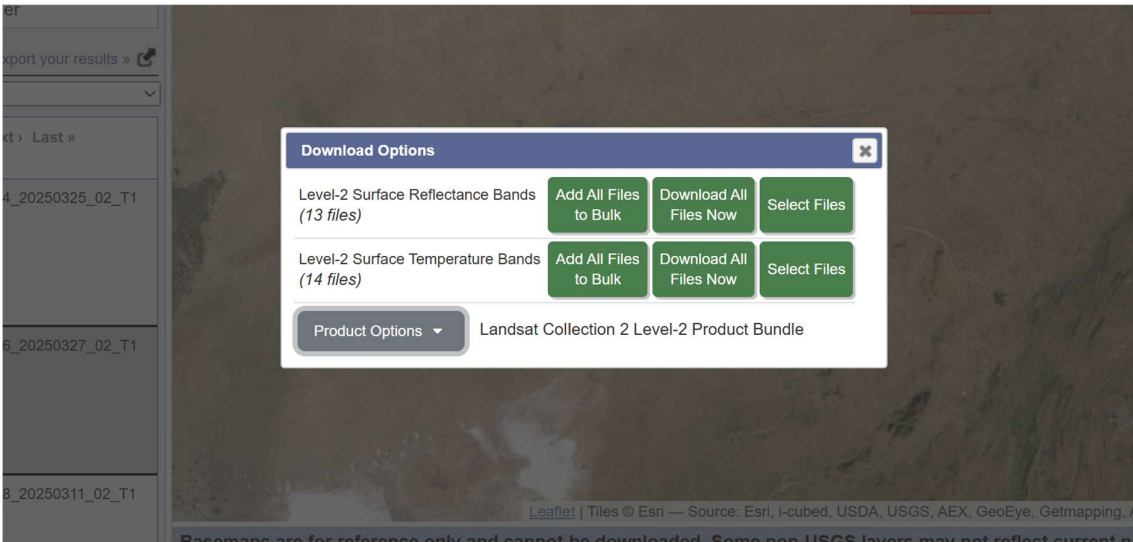


Figure 2-13: Download options available for Landsat Collection 2 Level-2 imagery

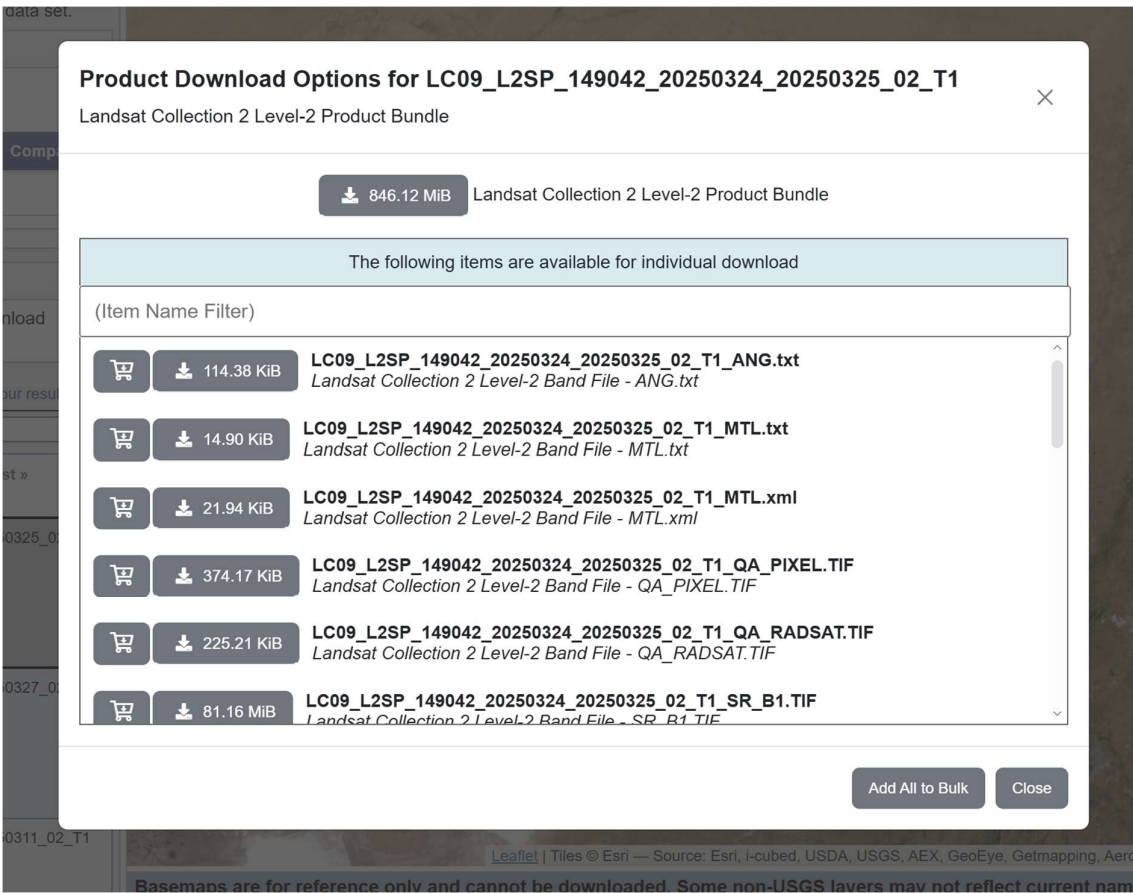


Figure 2-14: Product bundle containing multispectral bands and metadata files selected for download

2.7.2.9 Extraction and Organization of Data

The downloaded compressed file was extracted into the project directory. The extracted dataset contained multiple GeoTIFF files corresponding to different spectral bands along with a metadata file (MTL.txt). These files were organized for subsequent processing in QGIS and Python.

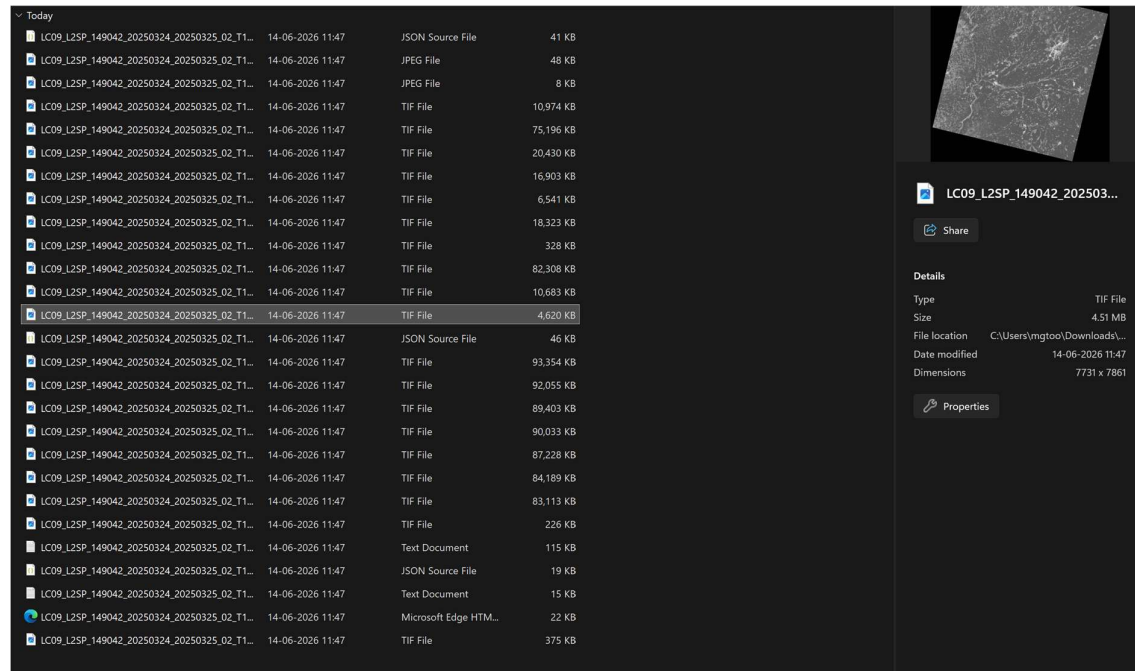


Figure 2-15: Extracted Landsat dataset showing multispectral band files and associated metadata

2.7.3 Data Download from Bhuvan Portal

Although Landsat imagery was used in this study, Bhuvan can also be utilized for obtaining Indian satellite datasets.

The general procedure includes:

- Accessing the Bhuvan portal.
- Selecting the desired dataset.
- Defining the study area.
- Downloading available imagery.

2.8 Landsat Dataset Structure

After downloading and extracting the Landsat dataset, multiple files become available. These files contain spectral information acquired by different bands of the Landsat 9 sensor.

2.8.1 Downloaded Dataset Components

A typical Landsat dataset contains:

- Spectral band files.

- Metadata files.
- Quality assessment files.

These files are stored in GeoTIFF format and can be imported into GIS and image processing software.

2.8.2 TIFF Files

The primary image data are stored as TIFF files.

Examples include:

- Band 1 (B1.TIF)
- Band 2 (B2.TIF)
- Band 3 (B3.TIF)
- Band 4 (B4.TIF)
- Band 5 (B5.TIF)
- Band 6 (B6.TIF)
- Band 7 (B7.TIF)

For the present study, Bands 2, 3, 4, and 5 are particularly important because they are used for RGB composite generation and NDVI computation.

2.8.3 Metadata File (MTL)

The metadata file contains important information regarding:

- Acquisition date.
- Sensor characteristics.
- Projection information.
- Radiometric parameters.
- Processing details.

2.8.3.1 Purpose

The metadata file helps users understand the characteristics and quality of the downloaded dataset.

2.8.3.2 Important Parameters

Important metadata parameters include:

- Satellite Name.
- Sensor Name.
- Acquisition Date.
- Path and Row.

- Spatial Resolution.

2.8.4 Organization of Project Data

For efficient analysis, the downloaded dataset should be organized into separate folders.

Recommended project structure:

Project Folder

- Raw_Data
- QGIS
- Python
- NDVI
- Classification
- Results

Proper organization facilitates data management and simplifies subsequent image processing tasks.

2.9 Chapter Summary

This chapter presented an overview of satellite data processing and introduced the major Earth observation missions used in remote sensing applications. The Indian Remote Sensing Programme, Landsat Programme, Sentinel Programme, and MODIS missions were discussed along with their characteristics and applications.

Particular emphasis was placed on Landsat 9 because it serves as the primary data source for the present study. The chapter also explained the concept of satellite resolution, including spatial, spectral, temporal, and radiometric resolution.

Furthermore, major satellite data sources such as USGS Earth Explorer and ISRO Bhuvan were introduced. The methodology used for downloading and organizing Landsat 9 imagery for the Jodhpur study area was described in detail.

The knowledge gained from this chapter provides the foundation for subsequent chapters involving QGIS-based image visualization, RGB composite generation, NDVI analysis, and machine learning-based image classification.

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